Q.1 Evaluation of \int\limits_0^1 {\int\limits_0^2 {{x^2}ydxdy} } is _____.

A. 4/3
B. 1/3
C. 2/3
D. 8/3

Q.2 What does the order of integration refer to in double integrals?

A. The type of region being integrated.
B. The sequence in which variables are integrated.
C. The size of the region of integration.
D. The number of iterations required.

Q.3 \eqalign{ & {\text{Let f be a function of two variables at a critical point}} (0,2), {f_{xx}}(0,2) = - 2, {f_{yy}}(0,2) = - 1 {\text{and}} {f_{xy}}(0,2) = 0 {\text{ }} \cr & {\text{so}} D = {f_{xx}}(0,2){f_{yy}}(0,2) - {f^2}_{xy}(0,2) = 2 \cr}

A. {{then}} {{f has a relative maximum at}} (0,2).
B. {{then no conclusion can be drawn}}{{.}}
C. {{then f has a saddle}} {{point at}} (0,2).
D. {{then f has a relative minimum at}} (0,2).

Q.4 The volume of parallelepiped with dimensions x, y, z and height h is V=xyh

A. True
B. False

Q.5 Which of the following equations represents a sphere in three-dimensional space?

A. {x^2} + {y^2} + {z^2} = {r^2}
B. {x^2} - {y^2} + z = {r^2}
C. {x^2} - {y^2} = {r^2}
D. {x^2} + {y^2} - 4y - {z^2} = {r}

Q.6 What is the condition of the extreme value theorem?

A. A function must be continuous and differential
B. A function must be continuous on a closed interval
C. A function must be differentiable
D. No conditions are necessary

Q.7 If the function f is continuous on the closed interval [a, b], then f has an absolute absolute minimum value on [a, b] maximum value and an

A. Rolle's Theorem
B. Mean Value Theorem
C. Extreme Valued Theorem
D. Squeeze Theorem

Q.8 Evaluation of \int\limits_0^1 {\int\limits_0^1 {dydx} } is _____.

A. 4
B. 2
C. 1
D. 0

Q.9 \int\limits_0^{\ln \,3} {\int\limits_0^{\ln \,5} {e^{u + v} \,du\,dv} } \,\, = \,\, - - - - - - - - -

A. 15

B. 5

C. 8

D. 3

Q.10 {\text{The}}\,\,{\text{absolute}}\,{\text{extrema}}\,\,{\text{of}}\,\,f(x) = {x^2} - 2\,\,{\text{is}}\,\,{\text{_____}}{\text{.}}

A. (0,\, - 2)

B. (0,\,2)

C. (0,\,0)

D. (0,\, - 1)
